

Telangana – Grade 10
2018 – Mathematics

Maximum Marks:40

1. Prathyusha stated that "the average of first 10 odd numbers is also 10". Do you agree with her? Justify your answer

Solution:

Odd numbers $\rightarrow 1, 3, 5, 7, 9, 11, 13, 15, 17, 19$

$$\begin{aligned}\text{Average} &= \frac{1+3+5+7+9+11+13+15+17+19}{10} \\ &= \frac{100}{10} = 10.\end{aligned}$$

2. Write the formula to find the median of a grouped data and explain the alphabet in it.

Solution:

$$\text{Median} = L + \frac{\frac{N}{2} - C_f}{f} \times h$$

$L \rightarrow$ Lower limit of median class

$N \rightarrow$ Number of observations

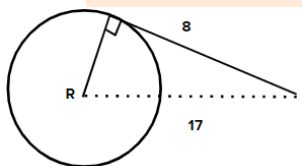
$C_f \rightarrow$ Previous cumulative frequency of median class

$f \rightarrow$ frequency of median class

$h \rightarrow$ class width

3. The length of the tangent to a circle from a point 17 cm from its centre is 8 cm. Find the radius of the circle.

Solution:



From tangent figure

$$R^2 + 8^2 = 17^2$$

$$R^2 = 289 - 64$$

$$= 225$$

$$R = 15\text{cm}$$

4. Find the value of $\tan 2A$, if $\cos 3A = \sin 45^\circ$.

Solution:

$$\cos 3A = \sin 45$$

$$\cos 3A = \cos(90 - 45)$$

$$\cos 3A = \cos 45$$

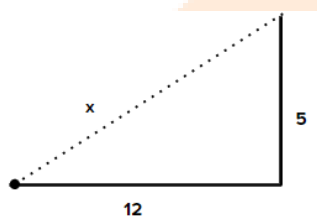
$$3A = 45$$

$$A = 15^\circ$$

$$\tan 2A = \tan 2 \times 15 = \tan 30 = \frac{1}{\sqrt{3}}$$

5. Srivani walks 12 m due East and turns left and walks another 5 m, how far is she from the place she started?

Solution:



According to direction figure,

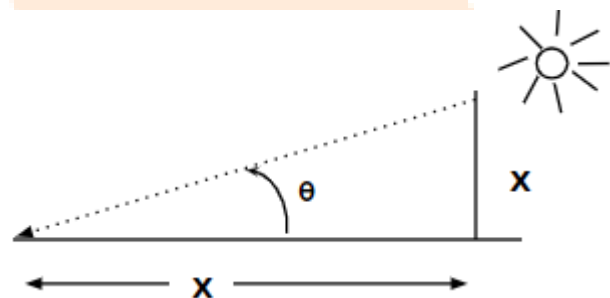
$$12^2 + 5^2 = x^2$$

$$x = \sqrt{144 + 25}$$

$$= 13\text{cm}$$

6. A pole and its shadow have same length, find the angle of the sun ray made with the earth at that time.

Solution:



$$\tan \theta = \frac{x}{x}$$

$$\theta = \tan^{-1}(+1)$$

$$= 45^\circ$$

7. What is the probability of getting exactly two heads, when three coins tossed simultaneously?

Solution:

Sample = $\{HHH, HHT, HTH, HTT, TTT, THH, TTH, HTT\} \rightarrow 8$

Even = $\{HHT, HTH, THH\} \rightarrow 3$

$$\text{Probability} = \frac{3}{8}$$

8. Find measure of the angles A and B, if $\cos(A - B) = \frac{\sqrt{3}}{2}$ and $\sin(A + B) = \frac{\sqrt{3}}{2}$.

Solution:

$$\cos(A - B) = \frac{\sqrt{3}}{2}$$

$$\cos(A - B) = \cos 30^\circ$$

$$A - B = 30 \quad (1)$$

$$\sin(A + B) = \frac{\sqrt{3}}{2}$$

$$\sin(A + B) = \sin 60^\circ$$

$$A + B = 60 \quad (2)$$

$$A - B = 30$$

$$A + B = 60$$

$$\hline 2A = 90^\circ$$

$$A = 45^\circ$$

$$B = A - 30$$

$$B = 45 - 30$$

$$= 15^\circ$$

9. What is the probability of a number picked from first twenty natural numbers is even composite number ?

Solution:

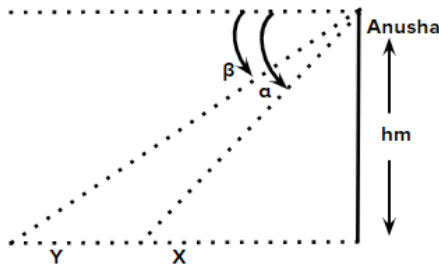
$$\text{Sample } n(s) = 20$$

$$\text{Event} = \{4, 6, 8, 10, 12, 14, 16, 18, 20\}$$

$$\text{Probability} = \frac{9}{20}$$

10. From the top of a tower of h m height, Anusha observes the angles of depression of two points X and Y on the same side of tower on the ground to be α and β . Draw the suitable figure for the given information.

Solution:



11. Find the median of $\frac{2}{3}, \frac{4}{5}, \frac{1}{2}, \frac{3}{4}, \frac{6}{5}$.

Solution:

$$Lcm = 60$$

$$\frac{2}{3} \times \frac{20}{20} = \frac{40}{60}, \frac{4}{5} \times \frac{12}{12} = \frac{48}{60}, \frac{1}{2} \times \frac{30}{30} = \frac{30}{60}, \frac{3}{4} \times \frac{15}{15} = \frac{45}{60}, \frac{6}{5} \times \frac{12}{12} = \frac{72}{60}$$

$$\text{Arrange} \rightarrow \frac{30}{60}, \frac{40}{60}, \frac{45}{60}, \frac{48}{60}, \frac{72}{60}$$

$$\text{Median} = \frac{45}{60} \rightarrow \frac{3}{4}$$

12. The height and the base radius of a Cone and a Cylinder are equal to the radius of a Sphere. Find the ratio of their volumes.

Solution:

Let radius of sphere be R

Height of cone = Radius of cone =

Height of cylinder = Radius of cylinder = R

Volume of cone = volume of cylinder = volume of sphere

$$\frac{1}{3} \pi r^2 h = \pi r^2 h = \frac{4}{3} \pi r^3$$

$$\frac{1}{3} R^3 = R^3 = \frac{4}{3} R^3$$

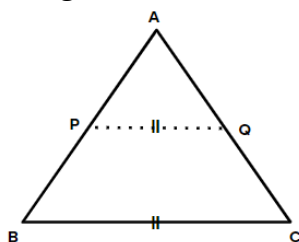
$$\frac{1}{3} : 1 : \frac{4}{3}$$

13. In $\triangle ABC$, $\overline{PQ} \parallel \overline{BC}$ and $AP = 3x - 19$, $PB = x - 5$, $AQ = x - 3$, $QC = 3cm$. Find x.

Solution:

$\therefore PQ \parallel BC$

Using B,P,T Theorem



$$\frac{AP}{PB} = \frac{AQ}{QC}$$

$$\frac{3x-19}{x-5} = \frac{x-3}{3}$$

$$9x-57 = x^2 - 3x - 5x + 15$$

$$0 = x^2 - 17x + 72$$

$$0 = x^2 - 8x - 9x + 72$$

$$= x(x-8) - 9(x-8)$$

$$= (x-8)(x-9)$$

$$x = 8 \text{ or } 9$$

14. How many silver coins of diameter 5 cm and thickness 4 mm have to be melted to prepare a cuboid of 12 cm x 11 cm x 5 cm dimension?

OR

Incomes of the families in a locality are given. Find the mode of the data.

Income (in Rs)	1-200	201-400	401-600	601-800	801-1000
Number of families	7	10	16	1	3

Solution:

(a)

$$\text{Volume of cuboid} = 12 \times 11 \times 5 \text{ cm}^3$$

$$\text{Volume of one coin} = \pi \times \left(\frac{5}{2}\right)^2 \times (0.4)$$

number

$$= \frac{7 \times 12 \times 11 \times 5 \times 4 \times 10}{22 \times 25 \times 0.4}$$

$$= 84$$

(b)

$$\text{Modal class} = 401 - 600$$

$$\text{Mode} = l + \left(\frac{f_m - f_1}{2f_m - f_1 - f_2} \right) \times h$$

$$l \rightarrow \text{lower boundary} = 401$$

$$h \rightarrow 200$$

$$f_m \rightarrow 16$$

$$f_1 \rightarrow 10$$

$$f_2 \rightarrow 12$$

$$\text{mode} = 401 + \left(\frac{16 - 10}{2 \times 16 - 10 - 12} \right) \times 200$$

$$= 401 + \frac{6}{10} \times 200$$

$$= 401 + 120 = 521$$

15. Prove that :

$$\frac{\cos A}{1 - \tan A} + \frac{\sin A}{1 - \cot A} = \sin A + \cos A.$$

OR

Show that:

$$(\sec \theta - \tan \theta)^2 = \frac{1 - \sin \theta}{1 + \sin \theta}.$$

Solution:

(a)

$$\frac{\cos A}{1 - \tan A} + \frac{\sin A}{1 - \cot A}$$

$$\frac{\cos A \times \cos A}{\cos A - \sin A} + \frac{\sin A \times \sin A}{\sin A - \cos A}$$

$$= \frac{\cos^2 A}{\cos A - \sin A} - \frac{\sin^2 A}{\cos A - \sin A}$$

$$= \frac{\cos^2 A - \sin^2 A}{\cos A - \sin A} = \frac{(\cos A + \sin A)(\cos A - \sin A)}{\cos A - \sin A}$$

$$= \cos A + \sin A$$

(b)

$$(\sec \theta - \tan \theta)^2$$

$$\frac{1}{\cos \theta} - \frac{\sin \theta}{\cos \theta} = \frac{(1 - \sin \theta)^2}{\cos^2 \theta}$$

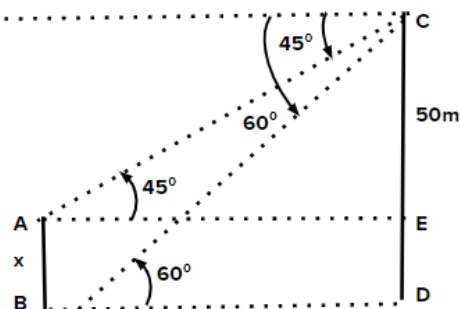
$$= \frac{(1 - \sin \theta)^2}{1 - \sin^2 \theta} = \frac{(1 - \sin \theta)^2}{(1 + \sin \theta)(1 - \sin \theta)} = \left(\frac{1 - \sin \theta}{1 + \sin \theta} \right)$$

16. Find the top of a tower of 50 m high, Neha observes the angles of depression of the top and foot of another building to be 45° and 60° respectively. Find the height of the building.

OR

From the deck of 52 cards, if a card is randomly chosen, find the probability of getting a card with (i) a prime number on it, (ii) face on it.

Solution:



Let height of building AB be x

$$\tan 45 = \frac{CE}{AE}$$

$$AE = 50 - x$$

$$\tan 60 = \frac{CD}{BD}$$

$$\therefore BD = AE = 50 - x$$

$$\sqrt{3} = \frac{50}{50 - x}$$

$$50 - x = \frac{50}{1.732}$$

$$50 - x = 28.87$$

$$x = 50 - 28.87$$

$$= 21.13m$$

(b)

$$n(s) = 52$$

(i) Event of pme number = $\{2, 3, 5, 7, 2, 3, 5, 7, 2, 3, 5, 7, 2, 3, 5, 7\}$

$$P(\text{prime}) = \frac{16}{52} = \frac{8}{26} = \frac{4}{13}$$

(ii) Event of face = $\{J, Q, K, J, Q, K, J, Q, K, J, Q, K\}$

$$P(\text{face}) = \frac{12}{52} = \frac{6}{26} = \frac{3}{13}$$

17. Construct an equilateral triangle XYZ of side 5 cm and construct another triangle similar to $\triangle XYZ$, such that each of its sides is $\frac{4}{5}$ of the sides of $\triangle XYZ$.

OR

Heights of the pupils of a particular school are given. Draw greater than cumulative curve and find the median height from it.

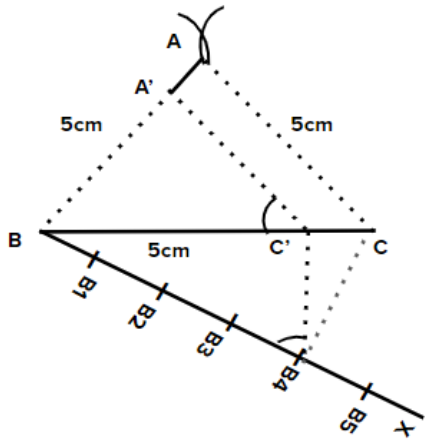
Height (in cm)	90-100	100-110	110-120	121-130	131-140	141-150
Number of pupils	5	2	3	8	8	6

Solution:

(a) Draw BC = 5 cm

Take 5 cm on compass radius

Draw $\triangle ABC$.



Draw BX at any acute angle

Taking equal fraction draw

B_1, B_2, B_3, B_4, B_5 .

Such that $BB_1 = B_1B_2 = B_2B_3 = B_3B_4 = B_4B_5$

Join B_4C and Draw B_5 parallel to B_4C .

Join $A'C'$, to get $A'BC'$.

(b)

Height	Greater	Number of pupils	Cumulative
90-100	Greater than 140	5	6
100-110	Greater than 130	2	14
110-120	Greater than 120	3	22
120-130	Greater than 110	8	25
130-140	Greater than 100	8	27
140-150	Greater than 90	6	32

